

Question:

A group of scientists is conducting an experiment on the human cells involved in the immune response. They genetically modify B cells so they do not express the cluster of differentiation 21 (CD21) on their cell surfaces. The pathogenesis of which of the following organisms would most likely be affected by this genetic modification?

Answer Choices:

- A. Epstein-Barr virus (EBV)
- B. Measles virus
- C. Human immunodeficiency virus (HIV)
- D. Human papillomavirus

Reasoning + Answer:

<think>

- CD 2 1 is a receptor for the complement component C 3 d , which is involved in signaling the B cell to undergo class -switching to produce Ig G antibodies .
- EB V and measles are both viruses that can bind to CD 2 1 .
- EB V specifically uses CD 2 1 as a receptor for entry into B cells .
- HIV uses CD 4 and C CR 5 or CX CR 4 , while human pap ill om avirus does not use CD 2 1 as a receptor .

- Given this information , the genetic modification of B cells to not express CD 2 1 would most affect the pathogenesis of EB V .
- Therefore , the answer is A .

</think><answer>A</answer>

